Information Systems Applications

Types of Information Systems and Their Applications

Information has become one of the primary factors in business success. We live in the Information Age, and those with the best, most current, most accurate information will win the day. Knowledge is power, and data is just another example of that power.

However, having vast amounts of good information is not enough; businesses and organisations need the best information systems to work with that information and produce the best outcomes.

Organisations need a solid information system in today's data-heavy world.

Six different types of information systems and how they are applied.

1. Business Support System

A BSS is a set of applications that have been unified under a global platform and whose objective is the management of processes and operations that support the functioning of an enterprise, company or business.

Business support systems (BSS) is the term traditionally used to describe the business and/or customer-facing functionality.

This software is used by employees and members of different areas of an organisation on a day-to-day basis to drive their activity.

BSS platforms are used by service providers, telecom operator delivery product management, customer management, revenue management (billing) and order management applications that help them run their business operations.

BSS helps service providers gain customer insight, compile real-time subscriptions and introduce new revenue generating services.

A good BSS provides system control and tailored scheduling to service providers, which facilitates the implementation of required changes without impacting customer satisfaction.

Use cases solved by BSS in the telecom sector

- ✓ Auto-Activation of SIM cards. Avoiding SIM Swapping and other frauds in the telecommunications sector.
- ✓ Landline, mobile, television and internet registrations.
- ✓ Collection, validation and submission of identification documentation for acceptance by the competent authorities of each market as required by the standard.
- ✓ Supervision and management of technical and operational activities of the entire proprietary and alternative commercial network.

- ✓ Comprehensive Point of Sale (POS) management with order, revenue, stock and back-office administration.
- ✓ Subscriptions and electronic notifications (certified communication) to customers.
- ✓ Intercommunication between teams, training of sales agents and gamification of sales incentives.
- ✓ Offer configurators.

Business support systems in other industries

While Business Support Systems (BSS) are fundamental and were born in the telecommunications industry, their usefulness extends to other industries. For example, utilities, such as electricity, gas, and water, use BSS for many purposes. In general, they work in the same way as in telecommunications, but they stand out:

- ✓ Manage customer billing and payments.
- ✓ Coordinate the distribution of services.
- ✓ Maintain accurate records of consumers and their consumption history.
- ✓ In the energy sector, BSSs play an important role in managing the billing of electricity and gas services and linking them to meter readings.
- ✓ Enable real-time monitoring of energy demand and generation.
- ✓ Facilitate contract management and service activation.
- ✓ Supply companies, such as food and commodity companies, use BSS to manage inventory and product orders.
- ✓ Optimize supply chains and reduce costs.

2. Decision Support System (DSS)

Decision Support System is a computer-based information system that combines models and data to solve semi-structured and some structured problems with intensive user involvement.

The DSS is a management-level, interactive, computer-based information system that helps managers make decisions. The Decision Support System gives middle managers the information necessary to make informed, intelligent decisions.

Decision Support Systems use different decision models to analyse or summarize large amounts of data into an easy-to-use form that makes it easier for managers to compare and analyse information. Often, these summaries take the form of charts and tables.

Example

A financial forecasting system that uses data analysis and modelling to assist company executives in making investment decisions by predicting future market trends.

3. Enterprise Resource Planning (ERP)

Enterprise resource planning is a software that integrates the planning, management, and use of all resources in the entire enterprise.

An ERP helps managers run the business from front to back. Departments can easily stay informed of what is going on in other departments that impact its operations or performance.

Being informed of potential problems and having the ability to work around them improve the company's business performance and customer relations.

Example

An ERP enables a manufacturer to share a common database of parts, products, production capacities, schedules, backorders, and trouble spots.

Responding quickly and correctly to materials shortages, a spike in customer demand, or other contingency is crucial because small initial problems are usually amplified down the line or over time.

4. Transaction Processing System (TPS)

Transaction processing is essential to helping businesses perform daily operations.

Transactions are any activity or event affecting the company and include deposits, withdrawals, shipping, billing customers, order entry, and order placement. TPS supports these business transactions.

Example

A point-of-sale (POS) system used in retail stores, where each sale is recorded and processed immediately, updating inventory levels and generating a receipt for the customer.

5. Office Automation System (OAS)

OAS comprises computers, communication-related technology, and personnel assigned to perform official tasks. It covers office transactions and supports official activity at every level in the organization, subdivided into managerial and clerical activities.

Office automation systems include the following applications:

- ✓ Email: The email application also covers file attachments such as audio, video, and documents.
- ✓ Voice Mail: This application records and stores phone messages in the system's memory, which can be retrieved anytime.
- ✓ Word Processing: Word processing covers the creation of documents, including memos, reports, letters, and anything else that's printable electronically. The

created text can be copied, edited, and stored via word processing commands, and it can be checked for grammar and spelling, line and word counting, and headers and footers.

Example

Microsoft Office Suite, where tools like Word, Excel, and Outlook help employees automate tasks such as document creation, data analysis, and email communication, improving overall productivity.

6. **Knowledge Work System (KWS)**

The KWS is a specialized system that expedites knowledge creation and ensures the business's technical skills and knowledge are correctly applied. The Knowledge Work System aids workers in creating and disseminating new information using graphics, communication, and document management tools.

Example

CAD (Computer-Aided Design) is used by software engineers and architects to design and test structures, enabling knowledge workers to create detailed models and simulations.

7. **Management Information System (MIS)**

Middle managers handle much of the administrative chores for day-to-day routines and performance monitoring, ensuring that all the work is aligned with the organization's needs. That is why MIS is such a valuable tool. Management Information Systems are designed to help middle managers and supervisors make decisions, plan, and control the workflow. The MIS pulls transactional data from various Transactional Processing Systems, compiles the information, and presents it in reports and displays.

Additionally, these reports can be produced monthly, quarterly, or annually, although MIS can have more immediate reports (e.g., hourly, daily).

Example

An HR management system that generates reports on employee attendance, performance, and payroll, helping managers make informed decisions about workforce management.

8. **Executive Support System (ESS)**

The ESS is like the MIS but for executive-level decision-making. Because the decisions involve company-wide matters, the stakes are higher, and they demand more insight and judgment.

The ESS provides greater telecommunication, better computing capabilities, and more efficient display options than the DSS. Executives use ESS to make effective decisions based on summarized internal data taken from DSS, MIS, and external

sources. In addition, executive support systems help monitor performances, track competitors, spot opportunities and forecast future trends.

Example

A dashboard system that provides CEOs with a real-time overview of key performance indicators (KPIs) such as sales revenue, market share, and customer satisfaction, enabling high-level decision-making and strategic planning.

Applications of Information Systems at Different Management Levels

1. **Top Management (Strategic Level)**

   Executive Support Systems (ESS): These systems provide high-level executives with real-time information and analysis to support strategic decision-making. They often include dashboards, data visualization tools, and predictive analytics.

   Enterprise Resource Planning (ERP): While ERP systems are primarily operational, top management can use them to gain insights into overall organizational performance, identify trends, and make strategic adjustments.

2. Middle Management (Tactical Level)

   Decision Support Systems (DSS): These systems help middle managers analyze data, evaluate alternatives, and make informed decisions. They often incorporate models, simulations, and what-if analysis.

   Management Information Systems (MIS): MIS provide middle managers with structured reports and information to monitor and control operations. They are essential for tactical decision-making and performance evaluation.

   Enterprise Resource Planning (ERP): Middle managers can use ERP systems to track key performance indicators (KPIs), identify bottlenecks, and optimize resource allocation.

3. Lower Management (Operational Level)

   Transaction Processing Systems (TPS): These systems handle routine, day-to-day transactions, such as sales, purchases, and inventory updates. They are essential for operational efficiency.

   Office Automation Systems (OAS): OAS tools like word processing, spreadsheets, and email help lower-level managers and staff streamline their tasks and improve productivity.

   Knowledge Work Systems (KWS): KWS support knowledge workers, such as engineers and designers, by providing tools for creating, sharing, and managing knowledge.

Specific Applications Across Different Systems

Business Support Systems (BSS)

Customer Relationship Management (CRM): BSS often include CRM systems to manage customer interactions, track sales opportunities, and improve customer satisfaction.

Order Management Systems: These systems handle the entire order-to-cash process, from order entry to invoicing and payment.

Enterprise Resource Planning (ERP)

Supply Chain Management (SCM): ERP systems often include SCM modules to optimize the flow of goods and services through the supply chain.

Human Resource Management (HRM): HRM modules help manage employee data, payroll, benefits, and performance evaluations.

Decision Support Systems (DSS)

Financial Forecasting: DSS can be used to predict future financial performance and make investment decisions.

Market Analysis: DSS can help analyze market trends, identify customer segments, and develop marketing strategies.

Office Automation Systems (OAS)

Document Management: OAS tools can be used to store, organize, and retrieve documents electronically.

Collaboration Tools: OAS often include tools like shared calendars and instant messaging to facilitate collaboration among team members.

Knowledge Work Systems (KWS)

Computer-Aided Design (CAD): CAD software is used by engineers and designers to create and modify product designs.

Content Management Systems (CMS): CMS are used to manage and publish digital content, such as websites and blogs.